

SAIFA Quant Edge 1.0 — Round 1

Risk Across Tails and Timescales

A wavelet–copula market-risk framework for multi-horizon tail dependence
Equal-weight US sector ETF portfolio (XLE, XLF, XLK, XLV, XLI, XLU, SPY)

Submission date: 7 October 2026

1. Motivation and research question

Market-risk engines usually fix one return horizon and one dependence structure (often a Gaussian correlation matrix). The academic wavelet–copula literature shows that co-movement—especially in the crash tail—varies with investment horizon and spikes differently in crises such as 2008 versus 2020. We ask: **does lower-tail dependence change with the investment horizon for a liquid US sector portfolio, and what does ignoring that do to measured VaR and Expected Shortfall (ES)?**

2. Data and design choices

Assets. Equal-weight basket of seven liquid SPDRs: energy (XLE), financials (XLF), technology (XLK), health care (XLV), industrials (XLI), utilities (XLU), and the broad market (SPY). Daily adjusted closes from Yahoo Finance, 4 Jan 1999–6 Oct 2026 (inner join). Log returns; portfolio return is the cross-sectional mean.

Margins. Per asset, GARCH(1,1) with Student-t innovations; standardized residuals mapped to uniforms by empirical PIT. **Wavelets.** MODWT (Daubechies db2, J=6) on residuals; horizon bands H1=D1+D2 (~2–8d), H2=D3+D4 (~8–32d), H3=D5+D6+S6 (~32–128d+). **Copulas.** Gaussian, Student-t, Clayton on each band; AIC selects the family. Primary 1-day risk uses the H1 AIC winner. **Risk.** Monte Carlo (N=10,000, seed=42) 1-day VaR/ES at 95% and 99%. **Benchmark.** Single-horizon Gaussian copula on the same GARCH residuals without wavelets. **OOS.** Estimation end 2019-12-31; evaluate 2020-01-02 onward with a 750-day rolling window, refit every 120 trading days (fast reproduction mode).

3. Does tail dependence change with horizon?

In-sample (≤ 2019), Clayton dominates AIC on every band. Estimated lower-tail dependence $\lambda_L = 2^{-1/\theta}$ is **high and persistent across horizons**: $\lambda_L(H1)=0.676$, $\lambda_L(H2)=0.686$, $\lambda_L(H3)=0.659$. Gaussian $\lambda_L=0$ by construction. Scale-to-scale differences are modest for this highly co-moving sector basket, but the **family choice is not**: a Gaussian surface forces zero asymptotic lower-tail dependence at every horizon. Figure 1 plots AIC-winner λ_L by band.

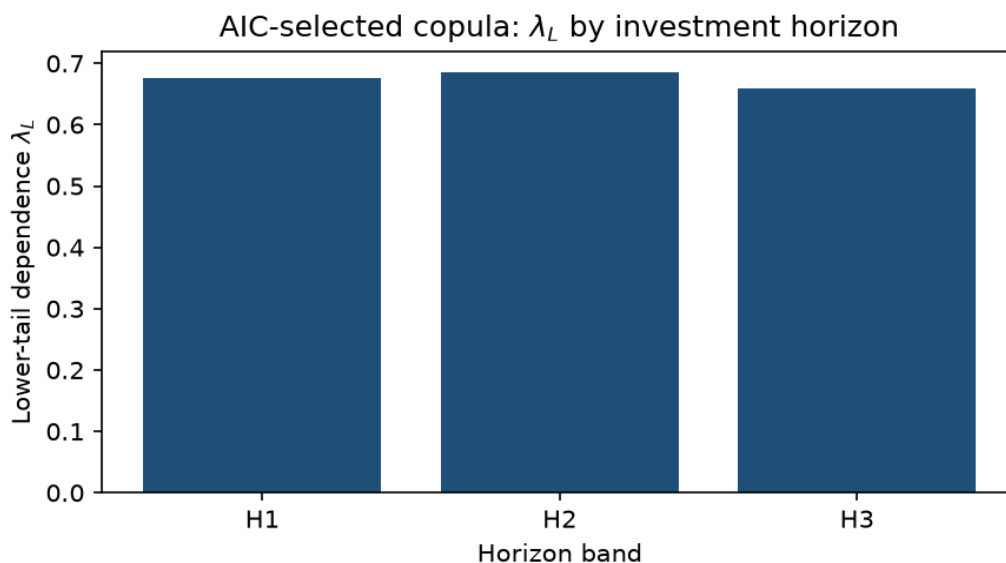


Figure 1. λ_L by horizon band (AIC winner).

4. Out-of-sample risk: ignoring tails understates risk

On 1,699 OOS days (2020–2026), the wavelet–copula (H1 Clayton) 95% VaR hit rate is **5.24%** (Kupiec $p=0.655$), close to the nominal 5%. The Gaussian benchmark hits **7.00%** (Kupiec $p=0.0003$), a clear under-coverage failure. Mean 99% ES is **0.0340** (WC) versus **0.0266** (Gaussian)—Gaussian understates crash ES by about **28%** on average. Both models show clustered exceptions (Christoffersen independence rejected), reflecting COVID and 2022 stress—honest evidence that static copulas still miss some dynamics.

Model	α	Hit %	Kupiec p	Mean VaR	Mean ES
Wavelet–copula	95%	5.24	0.655	0.0162	0.0229
Gaussian	95%	7.00	0.0003	0.0138	0.0187

Wavelet-copula	99%	2.35	1.80e-06	0.0268	0.0340
Gaussian	99%	3.18	6.51e-13	0.0215	0.0266

Table 1. OOS VaR/ES backtests (2020–2026).

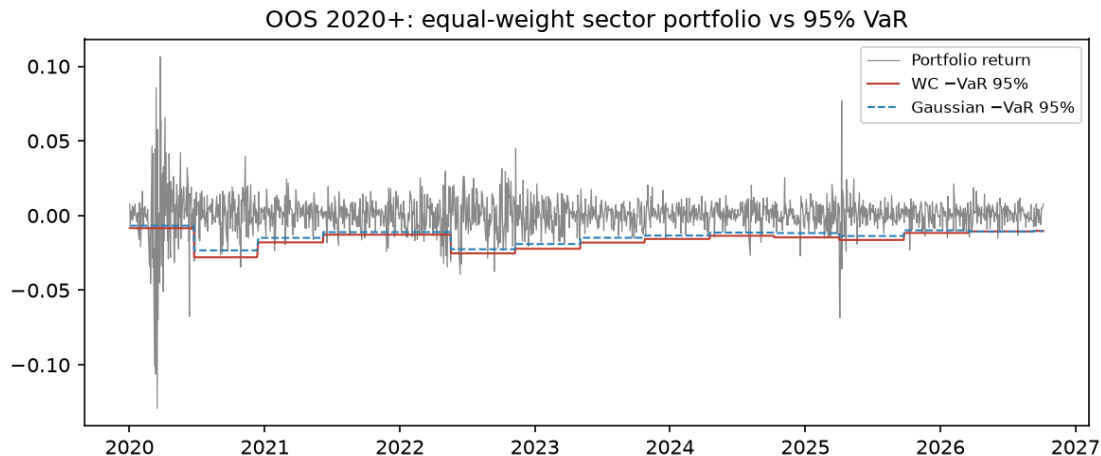


Figure 2. OOS portfolio returns vs -VaR 95%.

5. Answer and one manager recommendation

Answer. For this sector portfolio, lower-tail dependence is **material at every horizon** (Clayton $\lambda_L \approx 0.66\text{--}0.69$). Horizon-to-horizon variation is small relative to the Gaussian-vs-Clayton gap. **Ignoring tail dependence** (single-horizon Gaussian) **understates 99% ES by ~28%** and produces statistically rejected 95% VaR coverage in the COVID-inclusive OOS window, while the H1 wavelet-copula stays near the 5% target.

Recommendation (actionable tomorrow)

Desk / book: Sector ETF equal-weight prototype (XLE,XLF,XLK,XLV,XLI,XLU,SPY)

Regulatory LH anchor: 10d ES / internal 1d limit

Horizon diagnostic: $\lambda_L(H1)=0.676$, $\lambda_L(H2)=0.686$, $\lambda_L(H3)=0.659$; AIC winner H1: clayton

Stress contrast: OOS 2020–present 95% VaR hit rate wavelet-copula 5.24% vs Gaussian 7.00% (expected 5%); mean 99% ES WC=0.0340 vs Gaussian=0.0266

Action: Tighten tactical limits and size hedges to the H1 band; elevated coarse-horizon tail dependence also flags strategic diversification for SA high-p review.

Governance note: Wavelet-copula is an internal overlay; FRTB capital remains HS / MAR33 ES until validated.

6. Limitations

Static copulas miss exception clustering; vines and time-varying SJC were out of scope for Round 1. Fast reproduction uses 120-day refits and pairwise Clayton composite likelihood (not full 7-d MLE). Results are for liquid US ETFs and may not transfer to thin local markets. Wavelet-copula remains an **internal overlay**; FRTB capital stays on validated HS/ES engines.

7. Reproducibility

From the repository root: `make reproduce` (or `QUANT_EDGE_FAST=1 .venv/bin/python -m src.run_all`). One command regenerates tables, figures, and this PDF's numerical inputs under `outputs/`. Seed=42; data via yfinance with cached CSV.

References

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Christoffersen, P. (1998). Evaluating interval forecasts.
Jammazi, R., Reboredo, J.C. (2016). Energy — oil–stock wavelet-copula.
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Appendix A — AI use disclosure

AI assistants (Cursor / Composer) were used for literature synthesis coordination, code scaffolding, debugging, and report drafting. All methodological choices (assets, GARCH–MODWT–copula pipeline, OOS protocol, benchmark) follow the team’s locked design and the cited literature. Every number in this PDF is produced by `python -m src.run_all`; the team can explain, modify, and re-run the code.

Appendix B — Band AIC table (excerpt)

```
band, family, aic, loglik, lambda_l, lambda_u, p_rho_avg, p_nu, p_theta
H1, gaussian, -30550.810001051505, 15296.405000525752, 0.0, 0.0, 0.6078825413859291,,
H1, student, -30471.67493975247, 15257.837469876235, 0.009866390735206928, 0.009866390735206928, 0.6078825413859291, 30.0
,
H1, clayton, -43367.78384696412, 21684.89192348206, 0.6757008513252273, 0.0,,, 1.7682108233122067
H1, AIC_winner:clayton, -43367.78384696412, 21684.89192348206, 0.6757008513252273, 0.0,,,
H2, gaussian, -34415.447507195364, 17228.723753597682, 0.0, 0.0, 0.6074743981035787,,
H2, student, -34251.5999481834, 17147.7999740917, 0.009823583277067387, 0.009823583277067387, 0.6074743981035787, 30.0,
H2, clayton, -44929.75127206616, 22465.87563603308, 0.6859889838260865, 0.0,,, 1.8391051966774459
H2, AIC_winner:clayton, -44929.75127206616, 22465.87563603308, 0.6859889838260865, 0.0,,,
H3, gaussian, -32986.098584071195, 16514.049292035597, 0.0, 0.0, 0.5713717981641592,,
H3, student, -32706.476142773827, 16375.238071386913, 0.006669252781841393, 0.006669252781841393, 0.5713717981641592, 30.0,
0,
H3, clayton, -36615.48362438011, 18308.741812190055, 0.6587467760712941, 0.0,,, 1.6605665788874535
H3, AIC_winner:clayton, -36615.48362438011, 18308.741812190055, 0.6587467760712941, 0.0,,,
```